Answer **one** question in this section in the spaces provided.

8 **(a)** Define *magnetic flux density*.

.....

.....

..... [1]

(b) **(i)** Fig. 8.1 shows a flat circular coil carrying a current in the direction shown.

Sketch the magnetic field pattern due to the current in the coil within the dashed region.

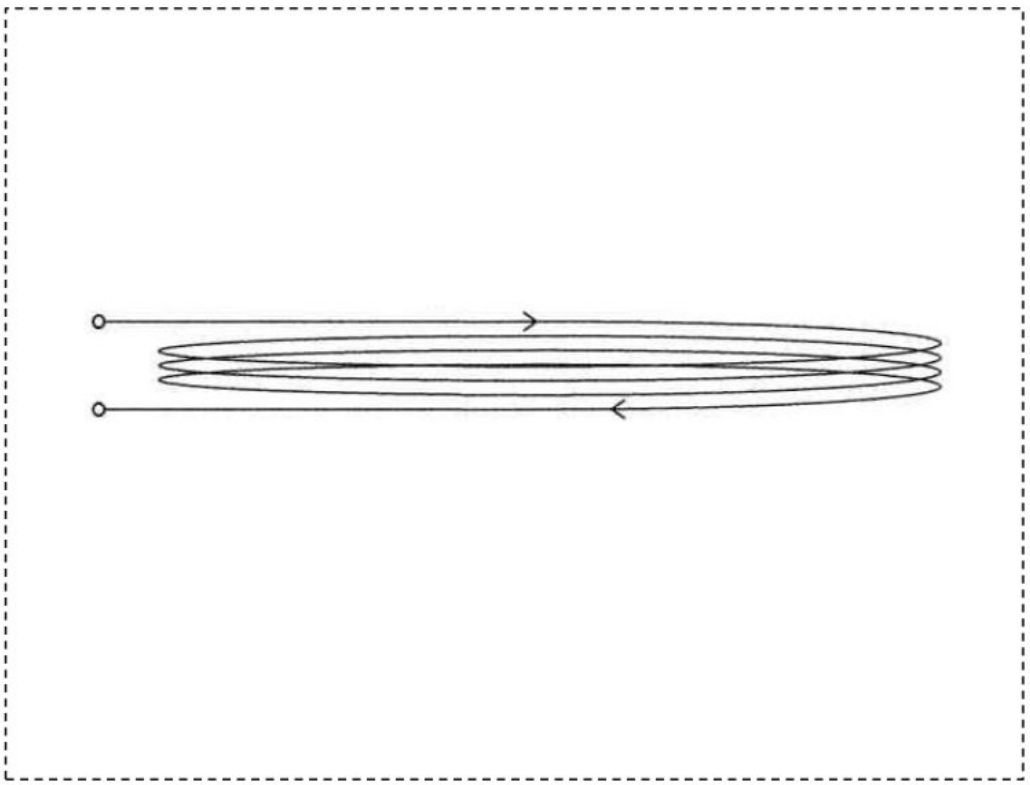


Fig. 8.1 (not to scale)

[3]

(ii) The flat circular coil in **(b)(i)** is fixed in position. A straight wire carrying current is then placed near the flat circular coil.

Explain the effect on the straight wire when it is placed

1. above the coil as shown in Fig. 8.2,

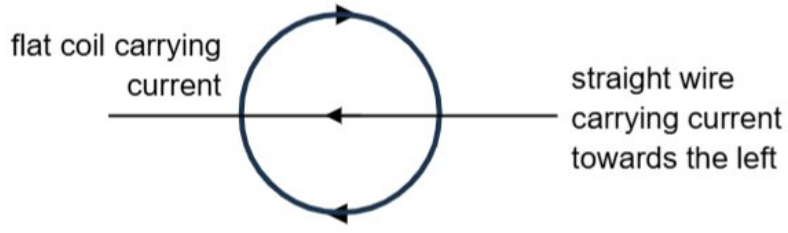


Fig. 8.2 (top view)

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2. along the axis of the coil, as shown in Fig 8.3.

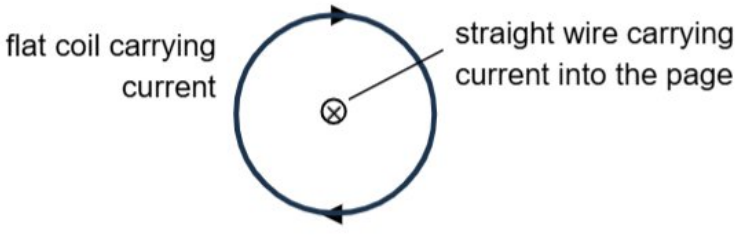


Fig. 8.3 (top view)

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.....

[3]

(c) Fig. 8.4 shows a metal saucepan placed on an electromagnetic induction cooking hob.

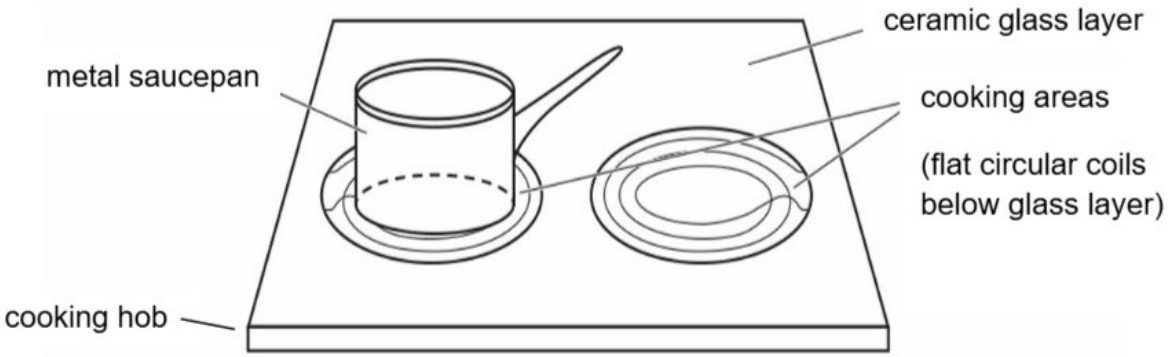


Fig. 8.4

Each cooking area of the induction cooking hob consists of a flat circular coil with 20 turns below a layer of ceramic glass. When switched on, the coils carry a high frequency alternating current.

(i) Explain how a high frequency alternating current in the coils results in a significant transfer of energy to the contents in the saucepan.

.....

.....

.....

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.....

.....

..... [3]

(ii) Fig. 8.5 shows the variation with time t of the alternating current I_{coil} flowing through the coil.

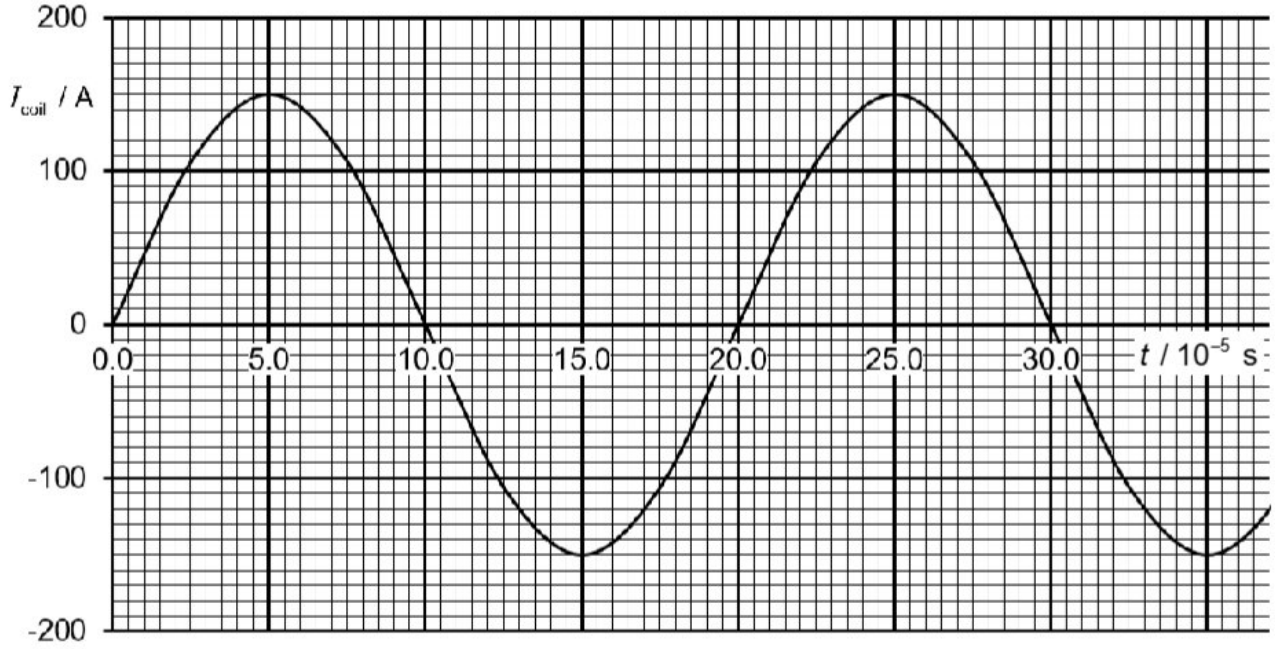


Fig. 8.5

The metal saucepan and its contents have a total mass of 900 g and an effective specific heat capacity of 2000 J kg^{-1} K^{-1}. The circular base of the saucepan has an effective area and resistance of 0.020 m^2 and 0.20 Ω respectively.

The magnetic flux density throughout the base of the saucepan can be taken to be $5\mu_o NI_{coil}$ where μ_o is the permeability of free space and N is the number of turns in the flat circular coil.

Assume there is no loss of energy between the coil and the saucepan.

1. Use Fig 8.5 to determine the maximum value of the induced current in the base of the saucepan.

maximum value of induced current = ... A [3]

2. Sketch, on Fig. 8.6, a graph to show the variation with t of the power dissipated in the saucepan for the first 30×10^{-5} s . Label all relevant values.

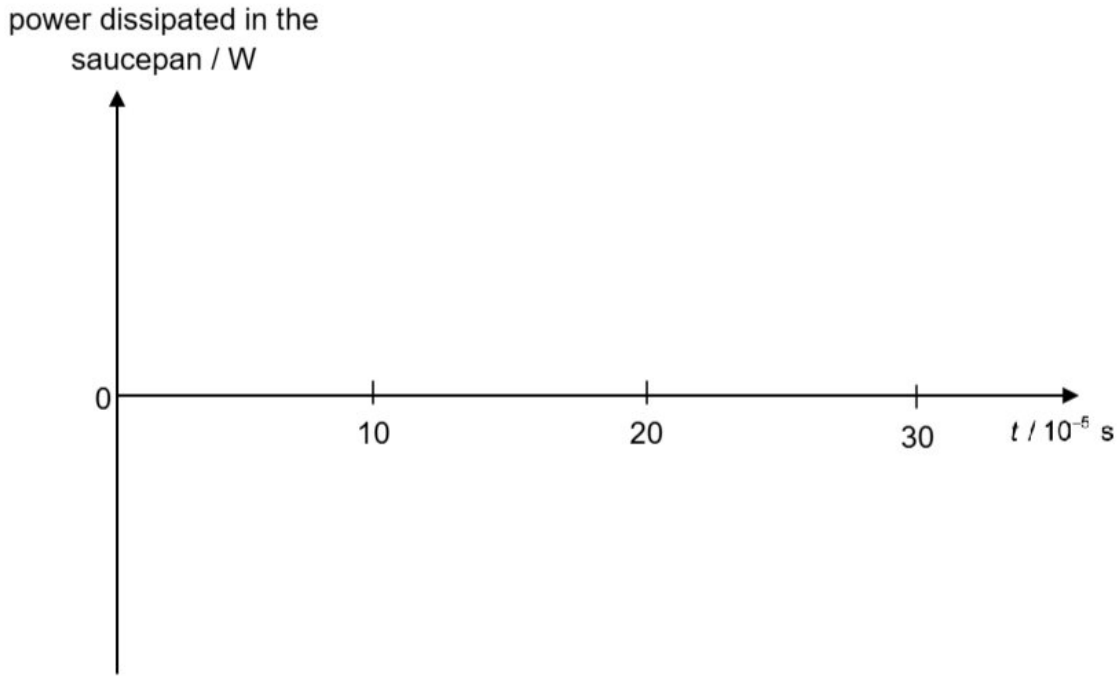


Fig. 8.6

[3]

3. Calculate the temperature rise of the saucepan and contents after 4.0 minutes of use.

temperature rise = ... ° C [3]

(iii) Stainless steel has a high magnetic permeability. Explain why stainless steel cookware are preferred in induction cooking.

.....

.....

..... [1]

[Total: 20]

9 **(a)** State how a stationary wave is formed.